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Group Contribution Method for Viscosities Based on Entropy Scaling Using the Perturbed-Chain Polar Statistical Associating Fluid Theory

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Supporting Information

ABSTRACT: In this work, we propose a new predictive entropy-scaling approach for Newtonian shear viscosities based on group contributions. The approach is based on Rosenfeld's original work [Rosenfeld, Y. *Phys. Rev. A* **1977**, *15*, 2545−2549]. The entropy scaling is formulated as third order polynomial in terms of the residual entropy as calculated from a group-contribution perturbed-chain polar statistical associating fluid theory (PCP-SAFT) equation of state. In this study, we analyze the course of entropy scaling parameters within homologous series and suggest suitable mixing rules for the parameters of functional groups. The viscosity of nonpolar, of polar, and of self-associating (hydrogen bonding) components are considered. In total, 22 functional groups are parametrized to viscosity data of 110 pure substances, from 12 different chemical families. The mean absolute relative deviations (MADs) to experimental viscosity data are typically around 5%. For three chemical families, namely, branched alkanes, 1-alcohols, and aldehydes, we obtain higher MADs of about 10%. Water is correlated with a MAD-value of 3.09%.

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INTRODUCTION

Detailed knowledge of the fluid viscosity is of importance in many engineering disciplines. High pressure applications like carbon capture and storage[1−3] or the design of new biofuels,[4−8] for example, often lack viscosity data. Furthermore, the design and optimization of solvents or working fluids relies on predictions of fluid viscosities.[9−13]

In many of those applications, the working fluid undergoes process steps in a wide range of temperature and pressure, involving phase transition. Due to the strong temperature and pressure dependence of the viscosity (see Figure 1), many measurements are needed for characterizing even pure substances by experiments. Predictive models, applicable to the complete range of conditions of technical relevance, are therefore desirable.[14]

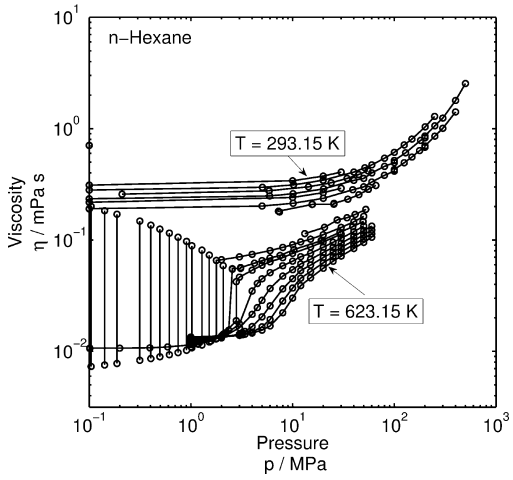

Figure 1. Experimental viscosities of *n*-hexane. Circles represent the experimental data; lines are a guide to the eye connecting viscosities at the same temperature.

Several promising approaches to model viscosities can be found in the literature. While some mainly focus on high accuracy for a specific component,[15,16] others focus on robust extrapolations or even predictions, for instance, Free-Volume Theory (FVT),[17−19] Friction Theory,[20,21] or the Dymond and Assael (DA) model[22−26] based on the Enskog equation.[27] An intriguingly elegant way of capturing the complex behavior of the viscosity is Rosenfeld's entropy-scaling,[28,29] where the viscosity is regarded as a function of the residual entropy, s_{res}, only. In his original work, Rosenfeld was the first to point out that, for spherical monatomic fluids, there exists a quasiuniversal, monovariable relation between a dimensionless form of the viscosity η^+ and the residual entropy. This means that the complex temperature and pressure dependence of the viscosity can be reduced to one simple function η^+ (s_{res}). This entropy scaling was rediscovered and confirmed from a more microscopic viewpoint by Dzugutov.[30]

Over the years, it was observed that the entropy scaling does not only hold for spherical monatomic fluids. For non-monoatomic fluids, the Rosenfeld-scaling does not obey a universal relation; for molecular substances, the entropy scaling leads to component-specific, but still monovariable, relations between the dimensionless viscosity and the residual entropy.[31−42] Using molecular simulations, Goel et al.[31] demonstrated a monovariable behavior for Lennard-Jones chains. This monovariable relation was described by a linear approach, introducing two scaling parameters. Goel et al.[31] showed that these scaling parameters can be correlated to the

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molecules chain length. Subsequent research on increasingly complex substances confirmed the monovariable relation for further *n*-alkanes,[33−35] for dumbbell-shaped particles,[38] and for water.[32,36,39] Limitations to the concept of transferable parameters were pointed out by Chopra et al.,[37] which suggests that the entropy-scaling as a monovariable relation between dimensionless viscosity and residual entropy is a powerful, but component-specific, concept. Excellent scaling with the residual entropy was confirmed for experimental viscosities of real systems.[40−42] Recent investigations revealed scaling invariances of dynamic and structural properties of many liquids along so-called isomorphs. Besides transport properties, also the residual entropy, s_{res}, was shown to be constant along an isomorph.[43−45] Therefore, isomorph theory might give new insights for the scaling of the viscosity with the residual entropy.

As pointed out by Rosenfeld[29] and others,[33,46,47] the dimensionless viscosity shows a steep increase in the low density region (i.e., $s_{res}/k_B \geq -0.5$), as shown in Figure 2.

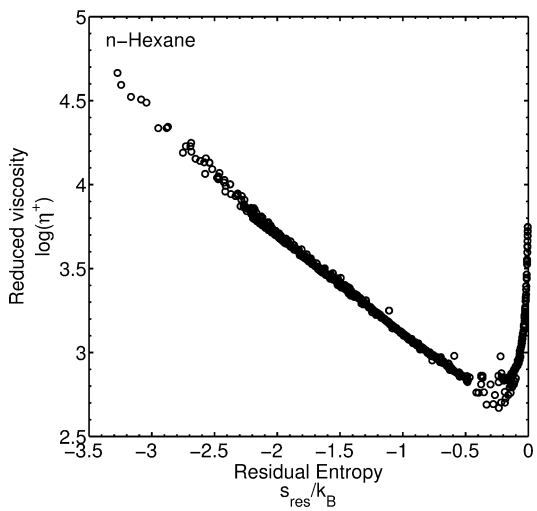

Figure 2. Reduced experimental viscosities of *n*-hexane according to Rosenfeld's approach,[28] i.e., eq 3.

Novak[46] proposed a different expression for the reduced viscosity by relating the viscosity of several *n*-alkanes to the corresponding Chapman-Enskog viscosity and obtained a nearly linear relation for the entire fluid region. This result yields the possibility to correlate and in particular extrapolate experimental (or simulation) data, so that already a small data set is sufficient to determine the viscosity, η, for the entire fluid phase region. Recently,[48] Novak modified his approach by additionally relating the reduced viscosity to entities, representing a predictive corresponding-states model for *n*-alkanes (and some further substances). Even though the "entities" have no direct structural correspondence, his results suggest that the scaling parameters can be correlated with the molecular structure.

In this work, we propose a new predictive approach for viscosities based on entropy-scaling. To calculate the residual entropy, we use a group contribution method based on the perturbed-chain polar statistical associating fluid theory equation of state (PCP-SAFT EoS) which was originally suggested by Vijande et al.[49] and was recently analyzed and reparameterized by Sauer et al.[50] The proposed model is entirely based on functional groups.

THEORETICAL BACKGROUND

Entropy Scaling. According to Rosenfeld's entropy-scaling approach, transport properties of pure substances are related to the molar residual entropy, $s_{res}(\rho, T) = (s(\rho, T) - s^{ig}(\rho, T))$, where ρ is the number density, T is the temperature, and the superscript ig indicates the ideal gas contribution. Rosenfeld's argument starts with a hard-sphere fluid (superscript hs), for which the viscosity η^{hs} is entirely determined by the dimensionless density $\xi^{hs} = (1/6)\pi\rho\sigma^3$, with σ as the particle diameter. Because the dimensionless density ξ^{hs} of a hard-sphere fluid is a monotone function of the residual entropy, this means that the viscosity of a purely repulsive fluid can be expressed as a unique function of the residual entropy, i.e.,

$$\eta^{hs}(\xi^{hs}(\rho)) = \eta^{hs}(s_{res}^{hs}(\rho)) \tag{1}$$

By using a first order perturbation theory, he pointed out that the residual entropy of a simple attractive fluid s_{res} can be approximated by the residual entropy of a hard-sphere fluid s_{res}^{hs} of the same dimensionless density. Rosenfeld then showed by simulations that the dimensionless viscosity η^+ of an attractive fluid is also entirely determined by the residual entropy s_{res}, so that

$$\eta^+(\rho, T) = \eta^+(s_{res}(\rho, T)) \tag{2}$$

The last step in this chain of arguments has no proof and is empirical, although statistical mechanical approaches such as mode coupling theory may provide arguments and assumptions on the validity for the last hypothesis. The dimensionless viscosity η^+ in Rosenfeld's original approach is obtained by dividing the viscosity by a reference viscosity η_{ref} given as

$$\eta^+(s_{res}(\rho, T)) = \frac{\eta}{\eta_{ref}} = \frac{\eta}{\rho^{2/3}\sqrt{Mk_BT/N_A}} \tag{3}$$

Applying this approach to experimental viscosity data of *n*-hexane (Figure 1) results in the expected monovariable behavior, Figure 2. The reduced viscosity is confirmed to be a function of the residual entropy, only. However, in the low density region (i.e., $s_{res}/k_B \geq -0.5$), the model shows a strong increase of the reduced viscosity due to the breaking down of the cubic lattice assumption as pointed out by Novak.[51]

It can be shown,[46] that the reference viscosity η_{ref} of Rosenfeld's approach is a simplified version of the Chapman-Enskog viscosity. As a first-order approximation of the Boltzmann-equation, the Chapman-Enskog viscosity is given as[52,53]

$$\eta_{CE} = \frac{5}{16}\frac{\sqrt{Mk_BT/(N_A\pi)}}{\sigma^2\Omega^{(2,2)*}} \tag{4}$$

Eq 4 expresses the pure component Chapman-Enskog viscosity as a function of the molar mass M, temperature T, characteristic particle diameter σ, and reduced collision integral $\Omega^{(2,2)*}$. The reduced collision integral is defined as

$$\Omega^{(2,2)*}(T^*) = \frac{\Omega^{(2,2)}}{\Omega^{(2,2)}_{hardsphere}} \tag{5}$$

with the reduced temperature $T^* = (k_BT)/\varepsilon$ and potential depth ε. The coefficients (2,2) are the sonine polynomial expansion coefficients and depend on the order of approximation of transport properties in the Chapman-Enskog formalism.[27,52]

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By approximating the volume per particle by a cube of edge length σ (cubic lattice structure) and applying the hard-sphere model ($\Omega^{(2,2)*} = 1$) to eq 4, the reduced viscosity η^+ is obtained as given in eq 3.

A more well-behaved function η^+ (s_{res}) in the region of low densities can be obtained by using the Chapman-Enskog viscosity η_{CE} according to eq 4 as reference viscosity η_{ref}. In the context of a group-contribution (gc) approach, it is necessary to express eq 4 in terms of gc parameters. Similar to Novak,[51] we relate the Chapman-Enskog viscosity to PCP-SAFT segments, with

$$\eta_{CE,gc} = \frac{5}{16}\frac{\sqrt{Mk_BT/(m_{gc}N_A\pi)}}{\sigma_{gc}{}^2\Omega_{gc}^{(2,2)*}} \tag{6}$$

The index gc indicates pure component parameters that are calculated with the homosegmented group contribution method based on PCP-SAFT EoS which will be addressed in the next section. The reduced viscosity is then defined as

$$\eta^* = \frac{\eta}{\eta_{CE,gc}} \tag{7}$$

To calculate the reduced collision integral $\Omega_{gc}^{(2,2)*}$, we use the empirical approximation by Neufeld et al.[54]

Group Contribution PCP-SAFT EoS. Within the PCP-SAFT EoS, molecules are considered to be linear chains of tangentially bonded identical segments.[55,56] These molecules are described by the pure component parameters m_i, σ_i, and ε_i. The homosegmented group contribution method proposed by Vijande et al.[49] suggests mixing rules to calculate pure component parameters by averaging the group parameters m_α, σ_α, and ε_α. Tihic et al.[57] introduced first order and second order groups[58] for a set of mixing rules similar to those of Vijande et al.[49] A group contribution method with a new set of mixing rules was proposed by Tamouza et al.[59] Besides homosegmented gc approaches, there exist heterosegmented gc approaches that consider molecules as chain molecules composed of nonidentical segments. Two important representatives of the heterosegmented approach are the SAFT-γ method by Lymperiadis et al.[60] and the gc-SAFT-VR method by Peng et al.[61] A heterosegmented gc approach based on the PC-SAFT EoS was presented by Paduszyński and Domańska,[62] Peters et al.,[63] and Sauer et al.[50]

An earlier study of our group showed the heterosegmented approach to be superior to the homosegmented model for phase equilibria. While liquid densities are described about equally well by both approaches, vapor pressures are described more accurately by the heterosegmented approach.[50] For the here considered case of predicting viscosities, the heterosegmented gc approach did not lead to significant improvements. Therefore, the pure component parameters from the combining rules by Vijande et al.[49] are used to calculate the residual entropy s_{res} as well as the Chapman-Enskog viscosity. Group parameters are taken from Sauer et al.[50] which have been adjusted to vapor pressure and saturated liquid density data. For methane, ethane, and methanol, we used the pure component parameters as published by Gross and Sadowski.[55,64] This group contribution method based on the PCP-SAFT EoS will henceforth be referred to as homosegmented GC-PCP-SAFT.

Viscosity Based on Group Contributions. In his original work, Rosenfeld[28] applied the principle of entropy scaling using a linear ansatz-function to describe the linear, monovariable behavior of dimensionless logarithmic transport coefficients (for the liquid phase region), according to

$$\ln \eta_i^+ = A_i + B_i\left(\frac{s_{res}}{k_B}\right) \tag{8}$$

Deviations from this linear relation were already reported in supercooled liquids,[38,65,66] alkanes at high densities,[37] and water.[39,42] In this work, we use a third order polynomial, where the second and third order terms account for deviations from the linear behavior that we found when applying eqs 7 and 8 to experimental viscosities. The empirical correlation function we use to determine the viscosity of a pure substance i is then

$$\ln \eta_i^* = A_i + B_i z + C_i z^2 + D_i z^3 \tag{9}$$

with

$$z = \left(\frac{s_{res}}{k_B m_{gc,i}}\right) \tag{10}$$

The division by $m_{gc,i}$ ensures that molecules of very different molecular mass have a similar range of values for the dimensionless entropy z. The viscosity parameters A_i to D_i of pure substances are obtained from the parameters A_α to D_α of functional groups α, respectively. We propose the following empirical relations for mixing group-contribution parameters

$$A_i = \sum_\alpha n_{\alpha,i} m_\alpha \sigma_\alpha{}^3 A_\alpha \tag{11}$$

$$B_i = \sum_\alpha \frac{n_{\alpha,i} m_\alpha \sigma_\alpha{}^3}{V_{tot,i}^\gamma} B_\alpha \tag{12}$$

$$C_i = \sum_\alpha n_{\alpha,i} C_\alpha \tag{13}$$

$$D_i = D \sum_\alpha n_{\alpha,i} \tag{14}$$

with

$$V_{tot,i} = \sum_\alpha n_{\alpha,i} m_\alpha \sigma_\alpha{}^3 \tag{15}$$

where $n_{\alpha,i}$ denotes the number of functional groups of type α in the substance i. The exponent γ and the parameter D were kept constant for all investigated substances and were optimized for n-alkanes. In summary, our approach for predicting viscosities applies a group contribution equation of state (GC-PCP-SAFT) with its own group contribution parameters. In order to predict viscosities, we adjusted group contribution parameters A_α, B_α, and C_α of all groups α, as well as model constants $D = -0.01245$ and $\gamma = 0.45$.

Procedure for Adjusting Model Parameters. The group contribution parameters related to the viscosity were adjusted to experimental data of pure substances. The experimental data was taken from the Dortmund Database[67] and is listed in the Supporting Information together with the temperature and pressure ranges and with references to the original publications.

Because the accuracy of the GC-PCP-SAFT EoS declines in the vicinity of the critical point, data points within the condition range

$$(0.9p_{crit} < p < 1.1p_{crit}) \wedge (0.9T_{crit} < T < 1.1T_{crit})$$

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were neglected.

The group contribution parameters related to the viscosity were determined with the following procedure: First, all available experimental viscosity data was read from data-files and the corresponding Chapman-Enskog viscosity and reduced viscosity of each data point were calculated. Then, the dimensionless residual entropy $z = s_{res}(k_B m_{gc,i})^{-1}$ was calculated from the GC-PCP-SAFT EoS using the analytical solution of

$$s_{res}(\rho, T) = -\left(\frac{\partial a_{res}}{\partial T}\right)_{\rho} \tag{16}$$

with the specific Helmholtz energy $a_{res} = A_{res}/N$ given by Gross and Sadowski.[55]

The functional group parameters of the mixing rules (eqs 11−13) were then optimized by minimizing the squared relative deviations of calculated to experimental viscosities for all data points of the considered substances using a Levenberg−Marquardt algorithm.[68]

■ RESULTS AND DISCUSSION

The homologous series of *n*-alkanes is important for developing a predictive approach based on group contributions. First, the effect of increasingly nonspherical molecular shape can be investigated, without superpositioning specific interactions due to (local) polar groups. Second, the PC-SAFT equation of state is well-suited to correlate and predict thermodynamic properties of *n*-alkanes. Third, there is comparably plentiful experimental data for *n*-alkanes. We first analyzed this homologous series in detail.

Parameter Behavior. The work of Novak[46] showed that the entropy scaling, with an appropriate ideal gas limit, leads to roughly linear scaling behavior of the logarithmic dimensionless viscosity and the residual entropy. In Figure 3 (symbols), we illustrate the scaling behavior of the viscosity for *n*-alkanes. In this study, we strive to approximately capture also the nonlinear part of the scaling behavior using a third order polynomial ansatz function. In order to propose suitable mixing rules for the group parameters (such as A_α for a group of type α) in eqs 11−14, we first adjusted the scaling parameters (such as A_i) individually for all *n*-alkanes i and analyzed how these parameters develop with increasing carbon number. The residual entropy s_{res} as well as the Chapman-Enskog viscosity was still calculated using the homosegmented GC-PCP-SAFT EoS.

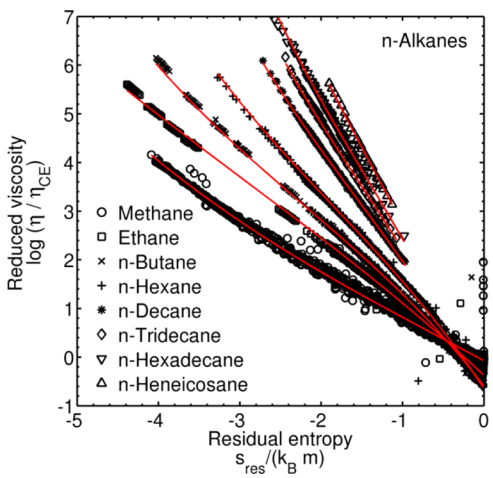

Figure 3. Reduced viscosity of a selection of the investigated *n*-alkanes. Symbols represent reduced experimental viscosities for residual entropies calculated with GC-PCP-SAFT, and red lines show the result of our correlation.

We observed that the correlation of parameters, especially of C_i and D_i, is strong, due to the relatively small deviation of the scaling behavior from a linear relation. For obtaining meaningful parameters C_i and D_i, experimental data need to cover a large portion of the range of residual entropy. The individually adjusted parameters depend strongly on the number of data points as well as the pressure and temperature range. For ensuring a robust group contribution concept, we reduced the number of adjustable parameters by defining the parameter D_i with a constant (i.e., group-independent) coefficient, according to eq 14. The rational for this choice was the following: first, we observed that the D_i values that were individually optimized for each substance increase with length of the *n*-alkane species. Second, we tentatively made the Ansatz $D_i = \Sigma_\alpha n_{\alpha,i} D_\alpha$, which represented the *n*-alkanes well with $D_{CH_3} = D_{CH_2}$, i.e., with the same coefficient for the CH_3 and the CH_2 functional groups. We then adjusted all coefficients A_α to D_α to our entire set of data and observed that the values of D_α scattered around the value of $D = -0.01245$ and we adopted this value as a model constant in eq 14.

To support this decision, we defined a parameter $(f_\alpha^{(3)}/f_\alpha^{(4)})$ that measures how the result of the model with constant D deteriorates compared with the model where D is adjusted for each group. Specifically, $f_\alpha^{(4)}$ is the objective function of a model where parameters A_α to D_α were adjusted to each group α, and $f_\alpha^{(3)}$ is the objective function of the model where parameters A_α to C_α are adjusted and parameter D is set to a constant value of $D = D_\alpha = -0.01245$. Figure 4 illustrates this measure for the chemical families considered in this study. The diagram shows that $f_\alpha^{(3)}/f_\alpha^{(4)}$ is close to unity (always smaller than 1.3) for all cases, which means that setting D to a constant value only mildly weakens the regression result. The dashed line in Figure 4 represents the analogous measure, $(f_\alpha^{(3,0)}/f_\alpha^{(4)})$, but now for the case where $f_\alpha^{(3,0)}$ is the model with parameters A_α to C_α adjusted and parameter D is set to zero. It is shown that the objective function for this case is significantly higher and the correlation result is thus weaker compared to the proposed parametrization. We note that for ethers the objective function $f_\alpha^{(3,0)}$ is lower than $f_\alpha^{(4)}$. That is possible because the functional

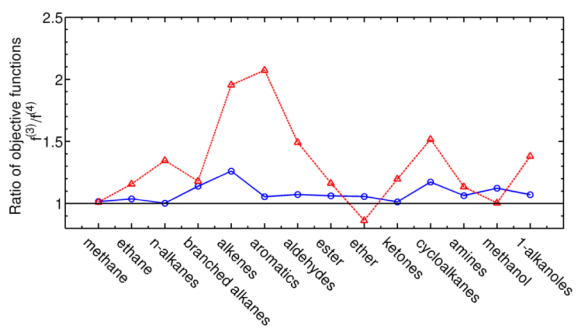

Figure 4. Ratio between the objective function of the proposed model $f_\alpha^{(3)}$ ($D_\alpha = -0.01245$) and the objective function of the same model, $f_\alpha^{(4)}$, but with 4 adjustable parameters, A_α to D_α (solid line with open circles). Furthermore, the ratio between the objective functions $f_\alpha^{(3,0)}$ and $f_\alpha^{(4)}$ is shown, where $f_\alpha^{(3,0)}$ is for a model with $D_\alpha = 0$ (dashed line with triangles).

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group parameters A_α to D_α were not adjusted simultaneously for all functional groups but successively in order as given in Figure 4.

For devising a robust group-contribution approach, it is important to verify that the viscosity parameters according to the mixing rules, eqs 11−12, are well-behaved. To evaluate the proposed mixing rules, we examine the behavior of the viscosity correlation parameters A_i and B_i with increasing number of carbon atoms. The basis of this consideration is a set of parameters A_i and B_i individually optimized for *n*-alkanes (with $D = -0.01245$ in eq 14). For comparison, this analysis was also performed for the case $C_i = 0$ and $D_i = 0$, which corresponds to eq 8. The resulting parameter behavior is shown in Figure 5.

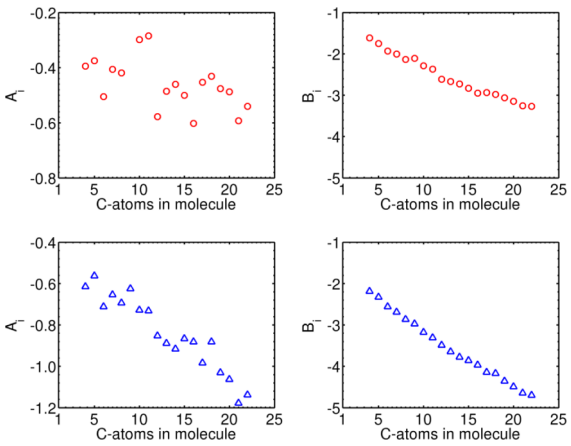

Figure 5. Viscosity correlation parameters A_i and B_i resulting from individual regressions to several *n*-alkanes. Reduced experimental viscosity was adjusted to either eq 8 (○) or eq 9 (Δ).

The slope-parameter B_i does scale very well with the chain length using either of the described correlation functions. The parameter A_i does not correlate well when using eq 8. Applying eq 9, the parameter A_i decreases monotonically with increasing chain length illustrating the suitability as a group contribution method.

Predicting the Viscosity of Substances Outside the Training Set. To assess the predictive ability of the group contribution method, we divided the available data set into a training set and an evaluation set. In a first step, we considered the homologous *n*-alkane series. Only the data of *n*-butane to *n*-nonane was used to determine the alkane viscosity group contribution parameters A_α−C_α (training set). Those parameters were then used to predict the viscosity of the remaining *n*-alkanes up to *n*-hexatriacontane (evaluation set). The resulting deviations of calculated to experimental viscosities are shown in Figure 6.

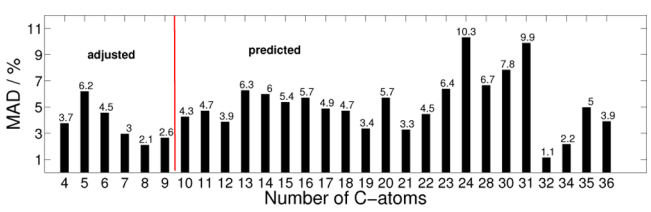

Figure 6. Deviations of the predicted viscosity of 22 *n*-alkanes with parameters adjusted to experimental data of six *n*-alkanes.

The deviations reported in Figure 6 are given as mean absolute relative deviations (MADs), MAD% = $(100/N_{data})\Sigma_1^{N_{data}}((|\eta_{calc} - \eta_{exp}|)/\eta_{exp})$, because the MAD %-values are more intuitive than squared deviations. Figure 6 shows that the deviations remain nearly constant for the complete range of substances. The viscosity of the longest investigated *n*-alkane, *n*-hexatriacontane, is predicted with a deviation of 3.9%, implying that the parameters of the training set can reliably be extrapolated. As an example, the predicted viscosity of *n*-octadecane as well as the experimental results from literature are shown in Figure 7.

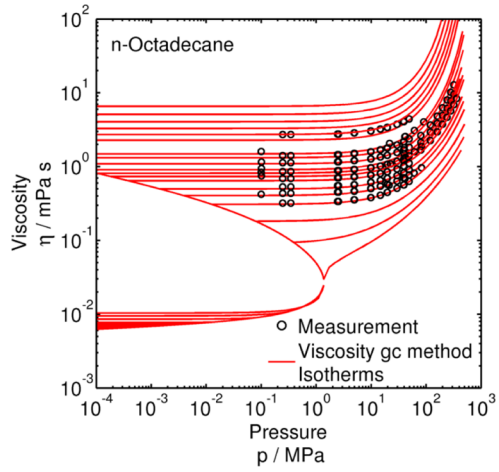

Figure 7. Predicted viscosity of *n*-octadecane.

The parameters adjusted to the six *n*-alkanes were then used to also predict the viscosity of amines. Out of the 12 amines (see Table 3 of the Supporting Information), only three primary and one secondary amine were used to determine the additional viscosity group contribution parameters of amines (training set). The resulting MADs are given in Figure 8.

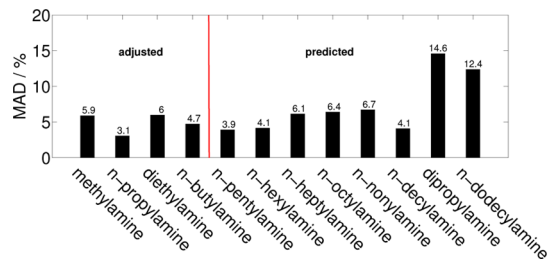

Figure 8. Deviations of the predicted viscosity of eight amines with parameters adjusted to experimental data of four amines.

Within the data set not considered for the parametrization (evaluation set), the deviations of the primary amines are only slightly higher than the correlated results in Table 4. The prediction for dipropylamine as a secondary amine is noticeable, because the MAD is three times higher than the correlated result. Given the small database in our training set, we still assess this result as sufficient. In summary, our analysis shows that the group-contribution method is suitable for predicting viscosities. More specifically, the parameters are well-behaved, so that the method can be applied to substances outside the training data set. We do not have sufficient data for substances with multiple functional groups to systematically assess predictions for such substances.

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Correlation Result for the Group-Contribution Method. For the final parametrization of the group contribution model, we included the data of all substances considered in our work. For the 31 *n*-alkanes, the parametrization resulted in a mean deviation MAD of 4.81%. Within the class of *n*-alkanes, the maximum deviation is 6.93%, which confirms the proposed mixing rules, eqs 11−13. Propane was omitted during parameter adjustment; therefore, the result for propane shown in Table 2 is predictive. The reduced viscosity $\eta^*(s_{res})$ of a selection of the investigated *n*-alkanes along with our calculation is shown in Figure 3.

Comparing Figures 1 and 3, the advantage of the entropy-scaling approach becomes obvious: The complex dependence of viscosity on pressure and temperature simplifies to a monovariable dependence of the viscosity on the residual entropy s_{res}. The substances mainly differ in slope and intercept, which corresponds to our parameters B_i and A_i, respectively. The slope as well as the intercept decrease with increasing chain length which is taken into account by our mixing rules. Note that if the Chapman-Enskog viscosity according to our parametrization, eq 6, were quantitatively correct for all fluids in the limit of ideal gases, then all intercepts would be zero and there would not be a (nonzero) A_i parameter. Figure 9 exemplarily shows the calculated viscosity $\eta(p,T)$ of *n*-octane applying the parameters in Table 1.

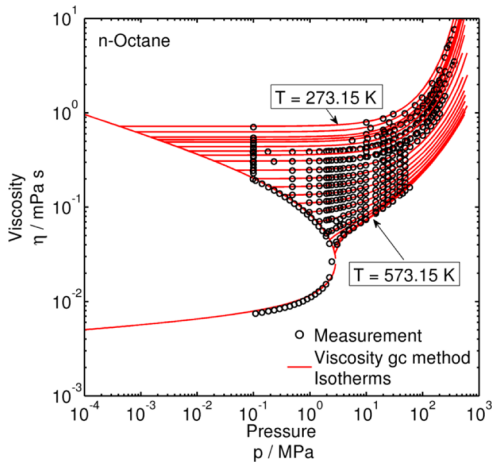

Figure 9. Calculated viscosities of *n*-octane using the parameters given in Table 1.

For branched alkanes, we obtain a mean deviation MAD of 10.1% with MADs for various branched alkanes ranging from 5.44% to 30.83%. The resulting MADs show the tendency to increase with decreasing number of available data points. In the case of 2-methylpentane and 3-methylpentane, our proposed approach can not differentiate between those two isomers. Both substances have little influence during the parameter regression and strongly depend on the quality of the experimental data: For instance, when the oldest of four references with experimental data of 2-methylpentane is omitted, we find a lower MAD-value of 13.44%. We note that other isomers differ with regard to the combined pure substance parameters. Examples are isomeric pairs of isopentane/neopentane; 2,2-dimethylbutane/2,3-dimethylbutane; and 2,2,4-trimethylpentane/2,3,4-trimethylpentane.

The investigated 1-alkenes and aromatics provide a good indication for the applicability of the alkane parameters to other classes of substances. Our correlation results (see Table 3) are in good agreement with experimental data with averaged MAD-values of 7.95% for 1-alkenes and 4.88% for aromatic substances, respectively. For both classes, the MADs are not increasing with increasing alkyl moiety. That suggests that the parameters obtained from the *n*-alkanes are well suited for these types of substances. Mesitylene and *p*-cymene show significantly higher MADs.

As polar substances, we investigated esters, ethers, aldehydes, and ketones. Overall, the calculated viscosity is in good agreement with the experimental data (see Tables 3 and 4). For ketones, the reduced viscosity is shown in Figure 10. Similar to alkanes, we find a nearly linear, monovariable dependence of the reduced viscosity on the residual entropy. Only methyl-isobutyl-ketone shows a curvature that is not captured by our correlation.

Ethenyl acetate is a good indicator for the transferability of the alkene parameters. Only 15 data points were available for

Table 1. Viscosity Group Parameters Obtained by Adjusting Eq 9 with Mixing Rules Eqs 11−13 to All Available Data of the Corresponding Group[a]

functional group α	$A_\alpha \times 10^3$	$B_\alpha \times 10$	$C_\alpha \times 10^2$
	Methane		
CH_4 (methane)	−1.5097	−1.0346	−1.7662
	Ethane		
$-CH_3$ (ethane)	−5.3556	−1.5252	−11.0450
	n-Alkanes		
$-CH_3$	−8.6878	−1.7951	−12.2359
$-CH_2-$	−0.9194	−1.3316	−4.2657
	Branched Alkanes		
−CH <	12.8159	−0.3416	5.5752
> C <	152.5629	12.3998	13.8406
	Alkenes		
$=CH_2$	−6.3736	42.4367	−10.8726
=CH−	−5.0637	−43.5678	−5.4247
	Aromatics		
−CH− (aromatic)	−4.7664	−1.6842	−6.5606
−C−R (aromatic)	14.4280	0.1104	−0.0699
	Aldehydes		
−R−CH=O	8.8675	−1.4782	−9.1325
	Ester		
−COO−	−0.3295	−1.1893	6.9576
	Ether		
$-OCH_3$	−4.6001	−1.6433	−5.3118
$-OCH_2-$	−5.5435	−2.2345	−9.9967
	Ketones		
>C=O	−2.2137	−1.4362	1.0417
	Cycloalkanes		
$-CH_2-$ (C5)	−3.8105	−1.6280	−6.7905
$-CH_2-$ (C6)	−3.7896	−1.4796	−4.5017
	Amines		
$-NH_2$	−4.4048	−0.6089	−8.9017
>NH	17.3783	−0.2465	−11.5122
	Methanol		
H_3C-OH (methanol)	−11.8859	−0.1253	13.4714
	1-Alkanoles		
−OH	−15.7583	−2.5654	−23.1537
	Water		
H_2O[b]	−14.7515	−1.8512	−29.0646

[a]All the members of a chemical group are listed in the Supporting Information. [b]Individually adjusted parameter $D_{H_2O} = -0.04059$.

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Table 2. Correlation Result for Viscosity Data: Mean Absolute Relative Deviations MAD of Viscosities for All *n*-Alkanes

substance	MAD/%	number of data points
n-alkanes	4.81	12010
methane	4.13	2551
ethane	4.45	2132
propane*	10.92	1190
n-butane	3.78	602
n-pentane	6.12	1186
n-hexane	4.55	894
n-heptane	3.41	473
n-octane	2.26	491
n-nonane	1.65	123
n-decane	2.35	624
n-undecane	2.81	184
n-dodecane	2.46	136
n-tridecane	3.40	84
n-tetradecane	4.36	82
n-pentadecane	3.81	160
n-hexadecane	4.15	107
n-heptadecane	3.93	276
n-octadecane	4.33	300
n-nonadecane	3.72	243
n-eicosane	5.36	29
n-heneicosane	3.82	30
n-docosane	5.15	24
n-tricosane	7.32	11
n-tetracosane	7.44	15
n-octacosane	6.03	14
n-triacontane	4.82	3
n-hentriacontane	6.93	5
n-dotriacontane	1.78	8
n-tetratriacontane	1.19	2
n-pentatriacontane	4.66	12
n-hexatriacontane	4.11	19

*Prediction.

Table 3. Correlation Result for Viscosity Data: Mean Absolute Relative Deviations MAD of Viscosities for Hydrocarbons (Other than *n*-Alkanes)

substance	MAD/%	number of data points
branched alkanes	10.10	1846
isobutane	8.88	549
isopentane	5.44	305
neopentane	16.65	68
2-methylpentane	29.81	21
3-methylpentane	7.21	7
2,2-dimethylbutane	30.83	19
2,3-dimethylbutane	7.34	6
3-ethylpentane	16.54	6
2,4-dimethylpentane	23.27	6
2,2,4-trimethylpentane	7.16	387
2,3,4-trimethylpentane	7.36	16
squalane	19.84	282
2,2,4,4,6,8,8-heptamethylnonane	5.46	174
alkenes	7.95	883
1-propene	12.41	276
1-hexene	6.51	254
1-heptene	4.54	107
1-octene	3.45	106
1-nonene	8.20	87
1-decene	7.16	53
aromatics	4.88	4251
benzene	7.06	990
toluene	4.31	1735
ethylbenzene	6.86	364
1,3-dimethyl-benzene	2.11	572
1,4-diemthyl-benzene	3.30	459
mesitylene	13.73	29
p-cymene	12.68	15
n-heptylbenzene	6.07	42
n-nonylbenzene	1.53	15
n-dodecylbenzene	6.06	30
cycloalkanes	3.85	1089
cyclopentane	1.86	306
cyclohexane	4.62	783

ethenyl acetate, so that its impact during the parameter adjustment step is small and the good result (i.e., MAD of 5.53%) suggests sufficient transferability of the group-contribution parameters.

As self-associating substances, we investigate amines and 1-alcohols. The reduced viscosities are shown in Figures 11 and 12, respectively. In both classes of substances, the entropy-scaling approach results in a monovariable behavior. While the behavior of the amines can still be considered nearly linear, the alcohols show a strong curvature. Especially for longer alcohols like 1-decanol, our approach can not represent the strong curvature because the A_i−C_i parameters are dominated by the alkane parameters. Given the limited number of adjustable parameters, we assess the overall representation of the 1-alcohols as still satisfactory. We identify the need for a wider range of experimental data for higher alcohols and amines (toward the gas phase and to highly dense liquid phases) in order to refine our model or the parametrization of the model.

In view of the pronounced curvature of the $\ln(\eta^*)$-s_{res}-relation for alcohols, it is interesting to consider water as a strongly associating substance. For water, it is possible to eliminate any ambiguity about the quality in which s_{res} is determined. For pure water, the IAPWS-95 model is a highly accurate equation of state,[69,70] which gives s_{res} from correlation of best experimental data. Figure 13 shows the reduced viscosity η^* of water versus the residual entropy, calculated with the IAPWS-95 equation of state. We limited the experimental database for the viscosity of water to literature from the year 1990 or more recent. From Figure 13, it is possible to conclude, first, that water obeys the principle of entropy-scaling over the entire range of fluid conditions, with good accuracy. Second, the entropy scaling shows marked deviations from a linear relation, with a strong positive curvature at high densities (i.e., low s_{res} values), as well as in the vicinity of the critical point, around $s_{res}/k_B = -1.9$. Inspired by the question of an anonymous reviewer, we investigated subcooled liquids (as shown in the Supporting Information) and find that the principle of entropy scaling holds for the metastable region (with pronounced curvature of the scaling relation for associating substances).

We now consider s_{res} of water as calculated from the PCP-SAFT model. Water has individual group-contribution parameters in the PCP-SAFT equation of state. We here use pure component parameters that were adjusted to vapor pressure data, to liquid density data, and to supercritical isotherms, as $m = 1.271$, $\sigma = 2.82$ Å, $\varepsilon/k = 281.94$ K,

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Table 4. Correlation Result for Viscosity Data: Mean Absolute Relative Deviations MAD of Viscosities for Substances Containing Oxygen or Amine Groups

substance	MAD/%	number of data points
ether	4.43	292
dimethyl ether	3.19	117
dibutyl ether	3.63	76
methyl-*tert*-butyl ether	9.44	6
dipentyl ether	10.24	16
dipropyl ether	5.51	77
ester	5.45	504
n-propyl acetate	6.45	105
n-butyl acetate	6.12	114
isopentyl acetate	7.84	78
ethenyl acetate	5.53	15
n-pentyl acetate	3.29	102
n-heptyl acetate	3.81	90
aldehydes	10.10	297
acetaldehyde	34.79	11
propanal	6.07	7
butanal	8.91	221
2-methyl-propanal	9.83	50
3-methyl-butyraldehyde	14.08	8
ketones	6.27	680
acetone	5.24	189
3-pentanone	5.39	116
methyl-isobutyl-ketone	13.41	68
4-heptanone	4.89	116
2-hexanone	4.00	86
2-octanone	7.86	105
amines	5.10	631
methylamine	5.82	107
diethylamine	6.35	105
propylamine	2.96	44
dipropylamine	4.54	25
butylamine	5.04	122
pentylamine	4.10	61
hexylamine	4.66	60
heptylamine	6.12	47
octylamine	4.91	31
nonylamine	4.75	14
decylamine	1.90	14
dodecylamine	10.08	1
1-alcohols	10.98	3070
methanol	5.87	1428
1-propanol	16.00	372
1-butanol	18.04	290
1-pentanol	16.68	64
1-hexanol	16.78	66
1-heptanol	15.57	16
1-octanol	14.34	153
1-nonanol	7.96	129
1-decanol	7.34	136
1-undecanol	12.44	194
1-dodecanol	25.64	187
1-tetradecanol	18.03	8
1-octadecanol	5.81	27
miscellaneous		
water	3.09	1406

μ = 1.855 D, and Q = 3.44 D with four association sites according to Scheme 4C as defined by Huang and Radosz[71] with association parameters ε^{AB}/k = 952.66 K and κ^{AB} = 4.35 × 10^{-3}. Comparison of the scaling behavior according to the

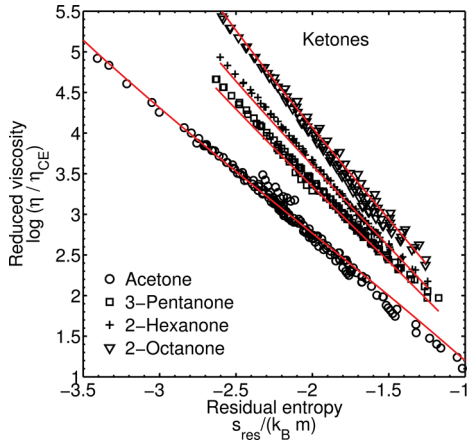

Figure 10. Reduced viscosity of the investigated ketones. Symbols represent reduced experimental viscosities for residual entropies calculated with GC-PCP-SAFT, and red lines show the result of our correlation.

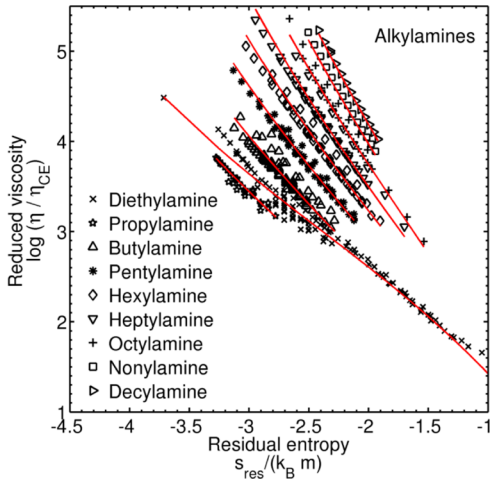

Figure 11. Reduced viscosity of amines. Symbols represent reduced experimental viscosities for residual entropies calculated with GC-PCP-SAFT, and red lines show the result of our correlation.

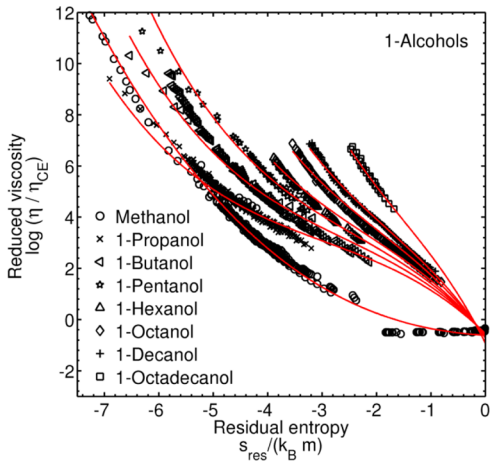

Figure 12. Reduced viscosity of 1-alcohols. Symbols represent reduced experimental viscosities for residual entropies calculated with GC-PCP-SAFT, and red lines show the result of our correlation.

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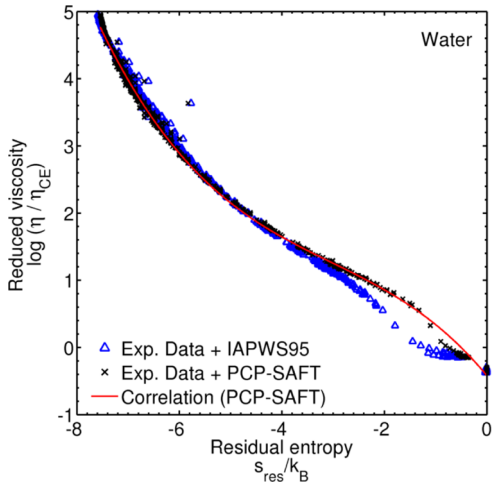

Figure 13. Reduced viscosity of water. Symbols represent reduced experimental viscosities versus residual entropies calculated with either the PCP-SAFT EoS (black crosses) or the IAPWS-95 formulation (blue triangles), respectively. The red line represents our correlation.

PCP-SAFT equation of state and the IAPWS-95 model, in Figure 13, shows good agreement in the dense phase, whereas for lower densities (i.e., $s_{res}/k_B > -2.5$), the results differ significantly. The most different s_{res} values from both equations of state occur in the vicinity of the critical point. Despite the different course of the data according to PCP-SAFT compared with IAPWS-95, we find a good, nearly monovariable scaling relation for PCP-SAFT. All four viscosity parameters A_α to D_α were adjusted for water. The correlation leads to average deviations (MAD) of 3.09% over the entire range of conditions, namely, 1406 data points, temperature from 248.15 to 715.57 K and pressure from 0.0522 to 60 MPa.

■ CONCLUSION

In this work, we present a new group contribution method for the correlation and prediction of pure component shear viscosities. The method applies a group contribution equation of state (PCP-SAFT) and uses the entropy scaling according to Rosenfeld. We formulate the entropy scaling in terms of three group contribution parameters per functional group. The method thus requires no pure component parameters. We observe entropy scaling to hold for strongly elongated substances, such as *n*-hexatriacontane and for hydrogen bonding species, such as 1-alcohols.

While the viscosity of linear molecules can be correlated and predicted very accurately, highly branched molecules show more significant deviations to experimental data. Members of the 1-alcohol family show the strongest hydrogen bonding of the substances we considered. For 1-alcohols, we observe pronounced deviations from a linear entropy scaling. Strongly hydrogen bonding fluids thus justify a further investigation. Moreover, our study is limited to substances with single functional groups. As a subsequent step, we evaluate the applicability of a group contribution approach to predict viscosities of mixtures and the prediction of other transport properties, such as thermal conductivity and diffusion.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.5b01698.

Discussion of the viscosity of subcooled liquids. Also temperature and pressure ranges as well as references to the original publications of all used experimental data (PDF)

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Notes

The authors declare no competing financial interest.

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